

Validating Parameters-classification

1.What is the percentage of total performance of random Forest and decision Tree?

- Random forest→Accuracy=0.92
- Decision Tree→Accuracy =0.90

2.What is the percentage correct classification of Non purchased in Random forest?

- The percentage of correct classification (Recall) of non purchased in Random forest is 0.94.

3. What is the percentage correct classification of purchased in Decision Tree?

- The percentage of correct classification (Recall) of purchased in Decision Tree is 0.83.
- $\text{Recall} = \frac{\text{True(Purchased)}}{\text{True(purchased)} + \text{False(unpurchased)}}$.

4.What is the percentage of correctly and wrongly classified of Purchased in random forest?

- The percentage of correctly and wrongly classified of Purchased (Precision) in random forest is 0.84.
- $\text{Precision} = \frac{\text{True(purchased)}}{\text{True(Purchased)} + \text{False(unpurchased)}}$.

5.How will you calculate the F1 measure?why?

- F1 measure talks about the overall performance and its calculated by
- $2 * \text{Recall} * \text{precision} / \text{recall} + \text{precision}$.

6.what is Macro Average?

- Macro average is the average of Recall, precision and f1 measure.
- It is calculated by $(\text{Recall(purchased)} + \text{Recall(unpurchased)}) / 2$ for recall and the same formula for precision and f1 measure .

7.In random forest, the average of performance of f1 measure in Macro average?

- In random forest,The average of performance of f1 measure in Macro average is 0.91.

8. In decision Tree, the average of performance of Recall in Macro average?

- In decision Tree, the average of performance of Recall in Macro average is 0.88.

9.what is weighted Average?

- The weighted average is the sum of proportion(weight) of the each class.

10. In Random forest, precision of non purchased in weighted average?

- In Random forest, precision of non purchased in weighted average is 0.92.

11. What is the support in classification reports?

- The support value is the total number of classes (purchased and non purchased) and total number of inputs test dataset.